

Does fiscal decentralization reduce energy poverty? Evidence from Vietnam

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ABSTRACT

This study examines the impact of fiscal decentralization on energy poverty in Vietnam, a country that has experienced significant change over recent decades, with fiscal policy reform, rapid economic growth, and expansion of energy access. We use household- and provincial-level data from 2004 to 2018 and employ different model specifications, including ordinary least squares (OLS) and instrumental variable (IV) estimates and two-step system generalized method of moments (GMM) estimates. Energy poverty is measured using a set of indicators based on income, expenditure, and the quality of energy consumption, while fiscal decentralization is calculated using both narrow and broad definitions of the share of revenue of local governments to total government revenue. We find that fiscal decentralization significantly reduces energy poverty. In particular, OLS estimates show that a one standard deviation increase in fiscal decentralization leads to average decreases in the probability of energy poverty of 0.3–1.3 percentage points at the household level and 0.3 to 1.6 percentage points at the provincial level. The IV estimates are consistent, and the GMM estimates also suggest long-run effects. Strong effects are found in lower-income households in the bottom three income quantiles and poorer regions. These results underline the potential use of fiscal decentralization as a policy tool for addressing energy poverty and emphasize the need for fiscal policy reforms that include the energy sector and account for regional disparities in economic activities.

1. Introduction

Energy poverty is an important challenge in sustainable development (Lu and Ren, 2023). Previous studies emphasize that energy poverty not only negatively impacts health and education (Katoch et al., 2024) but also has social, economic, and environmental consequences (Ayvar-Campos et al., 2019; Esen and Bayrak, 2017; Nguyen and Dinh Thanh, 2022). As such, scholars have paid great attention to the determinants of energy poverty in recent years.

However, there is still limited understanding of how fiscal policy affects energy poverty. Energy products and services, such as electricity or modern cooking tools, are basic needs in every household (Kalt et al., 2019). While governments are the main actors in addressing energy poverty (Aklin et al., 2018), countries seem to have different strategies for dealing with the issue. Bednar and Reames (2020) review the case of the US and find that federal-funded energy programs seem to be ineffective in reducing overall energy poverty. Kyprianou et al. (2019) show

that the EU has implemented programs and policies to alleviate energy poverty while funding projects to improve efficiency in energy consumption. Gu (2023) notes that Chinese governments provide energy subsidies to their citizens. The differences in the nature of energy poverty between developed and developing countries (Al Kez et al., 2024), as well as further variations brought about by climate and socioeconomic conditions (Awaworyi Churchill et al., 2022), highlight the need for a comprehensive understanding of the determinants of energy poverty.

Moreover, it is worth noting that energy poverty is a country-level problem with local characteristics. For instance, the national electricity grid is invested in by the central government to serve the whole country, so accessibility of the electricity supply is determined by central government policy. However, access to other energy services, such as clean fuels for cooking, or affordability of energy products and services, depends on local factors that include economic conditions and price levels. Thus, fiscal decentralization might affect energy poverty at the

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local level. More importantly, the literature suggests that fiscal decentralization has a significant impact not only on local governance (Dinh et al., 2023) but also on the drivers of local public goods delivery (Foa, 2022). Previous studies also note that fiscal decentralization has a mixed influence on local economic growth and development (Ary and Peduga, 2024), poverty (Wang and Deng, 2023), and income inequality (Hung and Thanh, 2022; Huynh et al., 2023). Further, Qurat-ul-Ann and Mirza (2020) carry out a meta-analysis of 30 studies on energy poverty. They report heterogeneity bias and highlight the need for subgroup-level analysis and consideration of local and regional factors in assessing energy poverty in developing countries. Therefore, it is important to understand how fiscal decentralization influences energy poverty, conditioned by local characteristics.

This study contributes to this line of literature by further examining the influence of fiscal decentralization on energy poverty in the case of a developing country. Using household- and provincial-level data for Vietnam from 2004 to 2018, this study investigates the impact of fiscal decentralization on energy poverty across multiple dimensions. These dimensions include financial constraints, affordability, access to modern appliances, and multidimensional energy deprivation. To address potential endogeneity, we employ plausible econometric methods, including instrumental variable (IV) estimation using regional variations in fiscal decentralization, and the two-step system generalized method of moments (GMM) to account for dynamic effects. The results document a significant negative relationship between fiscal decentralization and energy poverty. On average, a one standard deviation increase in fiscal decentralization corresponds to a 0.3 to 1.3 percentage-point reduction in the likelihood of household energy poverty and a 0.3 to 1.6 percentage-point reduction at the provincial level.

Our findings contribute to the literature on energy poverty in two ways. *First*, this study extends the literature on fiscal policy and energy poverty in a developing country. The literature suggests that, besides specific programs such as energy subsidies, it is essential to understand how changes in government policy, including fiscal policy, affect energy poverty. Among a small number of studies, Nguyen and Su (2022) reveal that government spending has a U-shaped influence on energy poverty in a sample of 56 developing countries. Xia et al. (2022) add that fiscal decentralization reduced energy poverty in China over the period 2005Q1–2019Q4. Wang et al. (2023) analyze the case of 30 Chinese provinces and conclude that fiscal decentralization reduces energy poverty, with residents gaining increased access to clean energy. Our study thus adds fresh evidence emphasizing that fiscal decentralization can alleviate energy poverty in a lower-middle-income country. Moreover, we carry out heterogeneity analyses and reveal that the effects are particularly significant among lower-income households in the bottom three income quantiles and economically disadvantaged regions, such as the Northeast and Northwest of Vietnam. These findings suggest that fiscal decentralization encourages local governments to address local energy needs and reduce disparities in energy poverty, which could have long-term benefits for equality.

Second, this study examines the impact of fiscal decentralization from two perspectives: a narrow definition, measured by the percentage of revenue fully retained by the provincial government, and a broad definition that includes shared revenue relative to the total expenditure assigned to the provincial government. Most previous studies seem to focus on one or other of these indicators, which might not encompass all aspects of fiscal decentralization. More interestingly, we also use an innovative, multidimensional measure of energy poverty (see Doan et al. (2025)). There has been enthusiastic debate about the definition of energy poverty, and it is agreed that it means there is insufficient access to energy and that energy products and services are unaffordable, thereby negatively affecting households' daily activities (Lu and Ren, 2023). For developing countries, scholars agree that the inaccessibility of modern energy products and services should be considered the main indicator of energy poverty, while the inability to afford these products and services should be the focus in developed countries (Al Kez et al., 2024;

Campagna et al., 2024). It can be seen from Fig. A1 (Appendix) that, except for low-income groups, the proportion of the population with access to electricity has been raised to nearly 100 % in all countries, especially high-income countries. This trend is also seen in Vietnam (see Fig. A2 - Appendix), implying that improvements in the electricity grid in developing countries over recent decades have eliminated problems with the accessibility of energy services, even in rural areas. There have also been improvements in other aspects of energy poverty in Vietnam, such as access to clean fuels and technologies for cooking (see Fig. A3 - Appendix). These changes mean that any examination of energy poverty in developing countries should include accessibility, affordability, and other aspects so as not to miss the existence of energy poverty from a multidimensional perspective (Pelz et al., 2018).

Recent studies seem to use only accessibility when measuring energy poverty in developing countries (e.g., Nguyen and Su (2022)), while a similar study to ours by Wang et al. (2023) applies a logit regression to estimate the probability of a household being energy poor. We construct a set of indicators to measure energy poverty in Vietnam following the study of Doan et al. (2025). We start by evaluating affordability in energy poverty measurement using the ten-percent rate (TPR) index, the half-median (M2) index, and the low-income high-cost (LIHC) index (see Herrero (2017) and Sy and Mokaddem (2022) for reviews). We then construct a multidimensional energy poverty index (MEPI), which is in line with current trends in measuring energy poverty in developing countries (e.g., see Laldjebaev and Hussain (2021), Omar and Hasanujjaman (2021), and Rizal et al. (2024)). Going further, we follow the study of Doan et al. (2025) to construct a new indicator of energy poverty based on a family's need for modern electrical facilities. While the TPR, M2, and LIHC indices compare household energy expenditure focused on the financial burden and affordability gaps, we also evaluate the quality of energy access using the limited modern technologies (MOT) indicator to measure whether households lack essential energy-consuming appliances.

The study is structured as follows. The next section reviews the literature. Section 3 describes the data and measurements in detail. Section 4 provides theoretical and empirical models. Empirical results are presented in Section 5. Section 6 discusses our findings, and the final section concludes this study.

2. Literature review

2.1. Energy poverty and its determinants: an overview

Energy poverty is an emerging field in energy economics (Katoch et al., 2024). In developed countries, energy poverty is defined as “*Energy poverty occurs when a household must reduce its energy consumption to a degree that negatively impacts the inhabitants' health and wellbeing*” (Commission, 2024). From a global perspective, the International Energy Agency (IEA) and the United Nations Development Programme (UNDP) suggest that energy poverty is the inability to access electricity, which is a more significant issue in developing countries (Tanaka et al., 2010). Generally, energy poverty can be understood as households facing “*insufficient energy access and unaffordability of modern energy*” (Lu and Ren, 2023). From these foundations, energy poverty is explored through two perspectives: accessibility and affordability (Simcock, 2020).

With the significant socioeconomic consequences of energy poverty (see Lu and Ren (2023)), scholars have also paid attention to its determinants. Generally, the literature suggests that energy poverty is driven by energy costs and income (Hills, 2011; Moore, 2012). Qurat-ul-Ann and Mirza (2020) review 30 empirical studies and find that energy price and income are the key determinants of energy poverty. Energy expenditure is one of the main costs in maintaining household living standards (Moore, 2012), so a decrease in income, especially when at the threshold of economic poverty, could push a household into energy poverty. Equally, an increase in energy costs, such as from rising

fuel prices, would push more households into energy poverty. Studying 32 African countries from 2005 to 2016, [Murshed and Ozturk \(2023\)](#) show that high energy prices are one of the main determinants of electricity inaccessibility.

Besides income and energy prices, the empirical literature has explored various potential determinants of energy poverty. Taking a multi-country perspective, recent studies have explored different determinants of energy poverty. [Nguyen et al. \(2021\)](#) suggest, using a sample of 65 economies from 2002 to 2015, that financial development can improve energy efficiency, reducing energy poverty. [Said \(2024\)](#) shows that education helped to erase energy poverty in 14 Latin American countries from 2000 to 2019. [Ongo Nkoa et al. \(2023\)](#) show that in 45 Sub-Saharan African countries from 1997 to 2018, countries with greater natural resource dependency appear to have higher levels of energy poverty.

In single-country studies, rich empirical literature identifies different determinants of energy poverty. On the one hand, previous studies have focused on overall energy poverty or regional differences in energy poverty. [Awaworyi Churchill et al. \(2022\)](#) show that cold temperature shocks increase energy poverty in Australia as they increase the need for energy for heating. [Yin et al. \(2023\)](#) add that financial inclusion reduces energy poverty in China. Recently, [Esquivel García and Toro-García \(2024\)](#) find that energy poverty, especially in municipal capitals, increased in Colombia from 2018 to 2022. They further add that remote areas with low population density experience high levels of energy poverty. On the other hand, scholars have examined energy poverty at the household level in individual countries. [Jayasinghe et al. \(2021\)](#) suggest that rural or single-person households, as well as those headed by females, younger adults, or individuals with lower levels of education, experience greater energy poverty in Sri Lanka. [Sokolowski et al. \(2020\)](#) suggest that rural households or households depending on retirement and disability pensions experience greater energy poverty in Poland. [Awan et al. \(2022\)](#) find that in Pakistan, households with lower levels of education, fewer resources, and female-headed families are more likely to be energy poor. Recently, [Manasi and Mukhopadhyay \(2024\)](#) reveal that more educated, small-sized, wealthier Indian households experience lower levels of energy poverty. [Rizal et al. \(2024\)](#) suggest that gender, business field, employment status, and education are the main determinants of energy poverty in Indonesian households.

2.2. Fiscal decentralization and energy poverty in Vietnam

There are studies on energy poverty in Vietnam, but none have considered the role of fiscal decentralization and its potentially significant impact on local government behavior and expenditure. [Kooijman-van Dijk and Clancy \(2010\)](#) focus on electricity accessibility by firms in Vietnam, Bolivia, and Tanzania, and conclude that poor accessibility is driven by firm location and electricity demand for firm services. [Son and Yoon \(2020\)](#) show that expanding a national electricity grid can increase electricity access, while household income is an important driver of electricity consumption. [Feeny et al. \(2021\)](#) find that temperature shocks from 2010 to 2016 led to energy poverty in Vietnam due to reduced agricultural output. Similarly, [Que et al. \(2022\)](#) add that temperature shocks induce energy poverty in Vietnam.

This study draws on the theoretical foundations of fiscal decentralization to examine its effects on energy poverty in Vietnam. Specifically, fiscal decentralization could promote greater efficiency in public service delivery and socioeconomic development, which could transmit to the reduction of energy poverty. Theoretically, fiscal decentralization is seen in the literature on public economics as effective in boosting governance quality in local government ([Plaček et al., 2020](#)). There are two current strands of literature on fiscal decentralization: the first-generation theory of decentralization (FGT), and the second-generation theory of decentralization (SGT) ([Dinh et al., 2023](#)). With FGT, fiscal decentralization is said to improve governance as it enhances fiscal competition between local governments ([Sepulveda and](#)

[Martinez-Vazquez, 2011](#)). Following the mechanism of “*voting with their feet*” in [Tiebout \(1956\)](#), under a decentralized system, local governments seek to meet citizens’ needs through improving the delivery of public goods and being accountable ([Oates, 1985](#); [Tiebout, 1956](#)). This approach is the so-called “*race to the top*” ([Tufail et al., 2023](#)). With SGT, the public sector is seen as a monopoly that seeks revenue maximization ([Oates, 2005](#)). Under fiscal decentralization, local governments have greater incentive to maximize their tax base to meet their responsibilities rather than focus on improving social welfare ([Weingast, 2009](#)). As a result, fiscal decentralization can lead to the problem of a “*race to the bottom*” ([Rodden, 2004](#)), where local governments compete in the opposite way, downgrading their quality of governance to attract income to the tax base ([Dong et al., 2012](#)). The empirical evidence is mixed (e.g., see [Altunbaş and Thornton \(2011\)](#), [van der Kamp et al. \(2017\)](#), and [Wang and Deng \(2023\)](#)).

In the case of Vietnam, there have been several studies on fiscal decentralization since the “Doi Moi” structural reforms in the 1980s ([Martinez-Vazquez and Gomez, 2005](#)). Many of these studies are discussion papers because of the limited data available for empirical analysis ([Bjornestad, 2009](#)). In recent years, more papers have been devoted to empirical analysis with favorable evidence of a positive influence of fiscal decentralization following FGT. For instance, [Dinh et al. \(2023\)](#) show that revenue decentralization improves public administrative procedures across regions in a sample of 63 provinces in Vietnam from 2011 to 2018. [Thanh and Canh \(2019\)](#) show that fiscal decentralization reduced public spending and the size of local governments in 63 Vietnamese provinces from 2006 to 2015. In terms of socio development, there is also positive evidence of the role of fiscal decentralization in promoting social welfare. For instance, [Nguyen \(2017\)](#) finds that fiscal decentralization positively affected provincial economic growth in Vietnam from 2004 to 2010. [Thanh and Canh \(2020\)](#) find that fiscal decentralization spurred economic growth in Vietnamese provinces from 2006 to 2015. [Diem and Hart \(2022\)](#) add that fiscal decentralization stimulates income convergence in Vietnam. [Do and Park \(2022\)](#) find that fiscal decentralization appears to increase foreign investment. [Dao and Hong \(2023\)](#) indicate that fiscal decentralization seems to improve health outcomes. Notably, [Bui et al. \(2023\)](#) reveal that, for a sample of 63 Vietnamese provinces from 2011 to 2019, fiscal decentralization appears to facilitate economic and social development with improvements in the efficiency of resource utilization. Thus, we propose that fiscal decentralization in Vietnam can stimulate the enhancement of local governance and so improve public service delivery, while the positive effect of fiscal decentralization on local governance can contribute to socioeconomic development. These benefits could then reduce energy poverty. Therefore, we propose the following hypothesis:

H: Fiscal decentralization can reduce energy poverty in Vietnam.

3. Data

3.1. The Vietnam Household Living Standards Survey

For household level data across provinces, we use the Vietnam Household Living Standards Survey (VHLSS) from 2004 to 2018. The dataset is collected by the General Statistics Office of Vietnam (GSO) with technical support from the World Bank and the UNDP. The VHLSS sampling process is conducted in three stages. Initially, around 3000 communes or wards are selected. Based on population data from the 1999 or 2009 censuses, these communes or wards are subdivided into enumeration areas. In the next stage, three enumeration areas are randomly chosen from each selected commune or ward to form the sample, which varied between survey waves. Lastly, two household samples are drawn from each enumeration area, with a short questionnaire sent to 15–20 households for the income survey and a long questionnaire sent to 3–5 households for the income-consumption survey ([Nguyen et al., 2019](#)).

This study uses data from the long household survey to focus on electricity spending for a given household income. The household sample comprises around 9000 households surveyed across eight waves (2004, 2006, 2008, 2010, 2012, 2014, 2016, and 2018), summing to 73,816 observations. Because VHLSS is designed as a rolling panel, i.e., around 50 % of the households surveyed in one round (e.g., VHLSS, 2014) are included again in the sample for the subsequent round (e.g., VHLSS, 2016), we are not able to harmonize a panel household dataset for a long period. Instead, in the further analysis, we collapse the data to the provincial level for panel data regression models.¹

Measures of energy poverty using income- and expenditure-based or quantity-based measures, each with distinct strengths and limitations, have been employed widely in recent studies (e.g., Lin and Wang (2020); Sadath and Acharya (2017); Doan et al. (2025)). The TPR index defines a household as energy-poor if it spends more than 10 % of its income on energy. This approach is simple to calculate and relies on widely available data. However, it has drawbacks, including potential errors in self-reported income and energy expenditure. Moreover, high-income households with high energy consumption may be misclassified as energy-poor, while the measure does not account for the energy efficiency of dwellings, which can significantly influence energy costs. To alleviate this drawback, we include another criterion to define a household as in energy poverty for this index; that is, if total household income is less than the median household income in the sample. So, the TPR index equals one if: (i) the share of electricity expenditure is greater than 10 % of the total household living costs, and (ii) total household income is less than the median.

A quantity-based approach to electricity poverty defines households as electricity poor if their per-capita electricity consumption is less than 100 kWh per year. This measure emphasizes the quantity of energy consumed to meet basic energy needs, especially in regions with underdeveloped infrastructure. However, it overlooks the proportion of income spent to reach this consumption level, potentially treating households as non-poor even when they meet the threshold by allocating a significant share of their income to energy. In Vietnam, progressive tariff rates on retail power are applied consistently across provinces. We follow Rademaekers et al. (2016) to define another energy poverty index based on median electricity spending (M2), where household electricity expenditure is less than half the median level of national household electricity expenditure.

Another widely used measure of energy poverty is the LIHC index, which considers a household to be energy poor if its energy costs exceed the national median and its residual income falls below the poverty line or 60 % of the national median income. This measure better identifies low-income households with high energy needs. However, its complexity and reliance on arbitrary thresholds make it harder to implement and interpret, particularly when accounting for different spending behaviors for living costs at the household level.

While the above measures based on quantity and income are commonly used and meaningful in evaluating energy poverty, evaluating high energy costs relative to income still ties energy poverty closely to financial circumstances. Given these limitations, a more contemporary approach should shift the focus from income and

expenditure to the quality of energy consumption (Feeny et al., 2021). This approach evaluates energy poverty based on the number and types of modern electrical appliances households use, as a direct measure of how energy access helps to improve living standards. By evaluating the availability and usage of appliances like refrigerators, washing machines, and air conditioners, this approach identifies both the quantity and utility of energy consumption. It also emphasizes inequality in energy access and the ability to benefit from modern technologies. For example, households with minimal use of appliances may have access to electricity but remain energy poor with limited benefits. We define a household as in energy poverty if it lacks modern electricity appliances (MOT).

Each of these indicators is binary, depending on whether or not a household is considered to be in energy poverty. If the household falls into the criteria for poverty, the indicator takes a value of 1 and is 0 otherwise. Finally, we employ the PCA approach to construct a multidimensional index for energy poverty (MEPI). Using the first component of each of TPR, M2, LIHC, and MOT, we compute a standardized index indicating the energy poverty of a household. This algorithm is effective in reducing a large set of variables while retaining most of the information from the original values (Ringnér, 2008). The value of MEPI is also normalized to 0–1.

Next, we collect a set of variables about household characteristics for our analysis. Before augmenting empirical model, in the next section, we present a theoretical framework, applying consumer theory, the household utility function, and decentralized public finance. The choice of household-level variables to explain energy poverty at the micro-level is related to previous models and empirical studies, such as Feeny et al. (2021) and Lin and Wang (2020). The baseline household-level variables in our analysis include socio-demographic characteristics, namely, age, gender, the education level of the household head, the number of household members, the size of the living area, and the number of electrical appliances. The variables also include a set of dummy variables indicating whether the household is an ethnic minority household, resides in an urban area, or lives in a rented house.

3.2. GSO data

For cross-province and panel analysis, we collect provincial-level data from the annual statistical yearbooks published by the GSO. Regarding the main predictor of energy poverty across provinces, we employ the level of fiscal decentralization in Vietnam, indicating how provincial governments manage their revenue and expenditure. According to the 2015 Vietnam State Budget Law, which is amended in 2020, provincial government revenue includes two components: fully retained revenue and shared revenue. Shared revenue is distributed to provinces based on a fixed rate, usually set for three to five years. This funding is designed to support local government fiscal management and planning over the medium term. During this medium-term period, local governments are incentivized to enhance their 100 % retained revenue as well as their shared revenue collected locally through taxes and fees. By promoting the generation of additional local revenue, fiscal decentralization aims to improve the financial autonomy of provincial governments and allow them to fulfill specific needs from their local spending.

Following Thanh and Canh (2020) and Dinh et al. (2023), we first define fiscal decentralization narrowly, taking the ratio of 100 % of provincial government retained revenue to total spending. As an alternative approach, we also calculate the ratio of 100 % retained revenue plus the shared revenue allocated to provincial governments relative to total spending, which we refer to as the broad definition of fiscal decentralization. Thus, fiscal decentralization is represented by two variables: fiscal decentralization by a narrow definition (FD1), and by a broad definition (FD2).

FD1 is computed as follows:

¹ From 2002, the VHLSS is conducted every two years to collect the information about living standards in Vietnam. The questions of the first wave in 2002 can be compared with prior surveys, such as the 1992–1993 Living Standards Survey (1993 VLSS), the 1993 Rich-Poor Status Survey (1993 RPSS), the 1997–1998 Living Standards Survey (1998 VLSS), and the Multipurpose Household Surveys (MPHS - conducted from 1994 to 1997 and in 1999). The VHLSS is developed by integrating the MPHS and VLSS surveys, with the 2002 survey being the first wave under the name VHLSS. Because many questions are revised in the early waves, we limit our sample to VHLSS 2004–2018 to ensure consistency. Also, most of the provincial-level data used to identify the impact of fiscal decentralization on energy poverty are available only after 2002.

$$FD1 = \frac{100\% \text{ retained revenue of provincial government}}{\text{total expenditure of provincial government}} \quad (1)$$

FD2 is computed as follows:

$$FD2 = \frac{100\% \text{ retained revenue} + \text{shared revenue from general government}}{\text{total expenditure of the provincial government}} \quad (2)$$

Fig. 1 shows cross-province heterogeneity in fiscal decentralization in Vietnam for 2004 and 2018. The figure distinguishes between two definitions of fiscal decentralization: the narrow definition, with a range of 0–0.6, and the broad definition, which extends the range from 0 to 1.² As depicted, there are changes in fiscal decentralization patterns over this 14-year period. In particular, the Red River Delta region shows an increase in fiscal decentralization, indicating that provincial governments in this region have gained more control over their revenue and spending. On the other hand, regions such as the Central Highlands and Mekong River Delta show a decrease in decentralization. The contrasting trends between regions might imply regional disparities in how fiscal decentralization policies are implemented and the varying impact on local public spending. Regions with higher levels of economic development, like the Red River Delta or Southeast regions, may benefit more from fiscal decentralization if resources are allocated efficiently to meet local needs. However, poorly performing economies in other regions may fail to meet revenue targets and be subject to reduced fiscal decentralization. Moreover, a decrease in overall fiscal decentralization at the national level also implies national policy changes with reforms for centralizing fiscal control.

We include important macro-level variables to explain socioeconomic patterns across provinces. These variables include provincial GDP growth,³ industrial output per capita measured by per-capita GDP in the industrial and construction sectors, population density, and inflation rate. We also include different sources of investment, namely, public investment, private investment, and foreign development investment (FDI), all calculated as thousand USD per capita.

Table 1 presents the summary statistics for the household and provincial-level variables used in the study. Access to the national power grid is nearly universal. However, over the study period, around one-quarter of the population faced energy poverty in Vietnam, with the TPR and MOT indices showing the highest averages. The average levels of fiscal decentralization, based on the narrow and broad definitions, are 0.128 and 0.196, respectively.

4. Theoretical and estimation models

4.1. Theoretical model

To model the linkage between fiscal decentralization and energy poverty, we apply consumer theory (Lancaster, 1966) augmented extensively to study household energy consumption (e.g., Ekholm et al. (2010); Li and Just (2018); Rodriguez-Alvarez et al. (2019) among others). We also use decentralized public finance theory (Musgrave, 1959; Oates, 1972; Tiebout, 1956). The modeling approach follows the standard individual model where a representative household derives utility from energy consumption and other consumption goods that are subject to income constraints and local public goods provision driven by fiscal decentralization policy.

We assume that household i in province j and year t obtains utility from a consumption bundle $U_{ijt} = U(E_{ijt}, C_{ijt})$ where E_{ijt} denotes the level

of energy consumed (e.g., electricity and clean cooking), and C_{ijt} represents the consumption of other necessary goods and services. Following consumer theory, the household seeks to maximize its utility given the budget constraint as:

$$\max_{E_{ijt}, C_{ijt}} \{U(E_{ijt}, C_{ijt}) \mid P_E E_{ijt} + P_C C_{ijt} \leq Y_{ijt}\} \quad (3)$$

where P_E and P_C are the prices of energy and non-energy goods, respectively, and Y_{ijt} is the household's disposable income.

The provision of local public goods and subsidies, including those affecting energy access, pricing, or infrastructure, is determined by the level of fiscal decentralization in province j and denoted as FD_{jt} . The parameter enters the model through two main channels by influencing the effective price of energy services $\tilde{P}_E$, and determining household income or transfers due to local fiscal policy.

To capture the energy poverty status of the household, we define an energy poverty index EP_{ijt} , which is a function of actual energy consumption, affordability, and access to modern energy sources. Thus, we specify the reduced-form equation of energy poverty as follows:

$$EP_{ijt} = f(FD_{jt}, X_{ijt}, P_{jt}, \lambda_j, \kappa_t) \quad (4)$$

where X_{ijt} is a vector of household-level characteristics (age, education, household size, etc.). P_{jt} includes provincial controls (infrastructure, economic development, etc.).

4.2. Estimation model

Based on the theoretical formulation, we postulate in our model specification that the probability of energy poverty occurring in a household is a function of the level of fiscal decentralization for local governments, in addition to other micro and macro covariates. The following regression model is estimated using ordinary least squares (OLS):

$$\text{Energy poverty}_{ijt} = \alpha + \text{Fiscal decentralization}_{jt}\beta + X_{ijt}\Gamma + P_{jt}\gamma + \lambda_j + \kappa_t + \varepsilon_{ijt} \quad (5)$$

where the probability of energy poverty $\text{Energy poverty}_{ijt}$ measures household energy poverty using one of the indices proposed above (TPR, M2, LIHC, MOT, and MEPI) for household i , in province j and year t . $\text{Fiscal decentralization}_{jt}$ is a variable denoting the level of fiscal decentralization in province j and year t . X_{ijt} is a set of control variables that account for household characteristics, including the age, gender, and education level of the household head, the number of household members, the number of electrical appliances, the living area size, and a set of dummy variables for minority status, urban residence, and rental housing status. P_{jt} is a set of control variables representing provincial characteristics, including provincial GDP growth, industrial output, population density, inflation rate, public investment, private investment, and FDI. λ_j and κ_t are dummy variables controlling for provincial heterogeneity and year-fixed effects. ε_{ijt} is the error term. The error term is clustered at the provincial level to account for heteroscedasticity over time within each province. The coefficient β represents the impact of fiscal decentralization on energy poverty.

Next, for panel data analysis, we average the energy poverty indicators at the provincial level and merge them with provincial-level data to employ panel data estimates. The following provincial-level panel model can be estimated using two-way fixed effects (2FE):

$$\text{Energy poverty}_{jt} = \text{Fiscal decentralization}_{jt}\delta + P_{jt}\gamma + \lambda_j + \kappa_t + \mu_{jt} \quad (6)$$

where the coefficient δ indicates the impact of fiscal decentralization on energy poverty at the provincial level. An obstacle in evaluating the impact of fiscal decentralization on energy poverty is endogeneity in the estimation model. Unobserved variables may correlate with the levels of fiscal decentralization. In Equation (5), unobserved factors ε_{ijt} include

² Appendix Figure A4 shows the distribution of fiscal decentralization levels using the two definitions, FD1 and FD2.

³ As energy poverty is a function of household income, we do not control for provincial total GDP or GDP per capita because these may introduce another source of confounding factors.

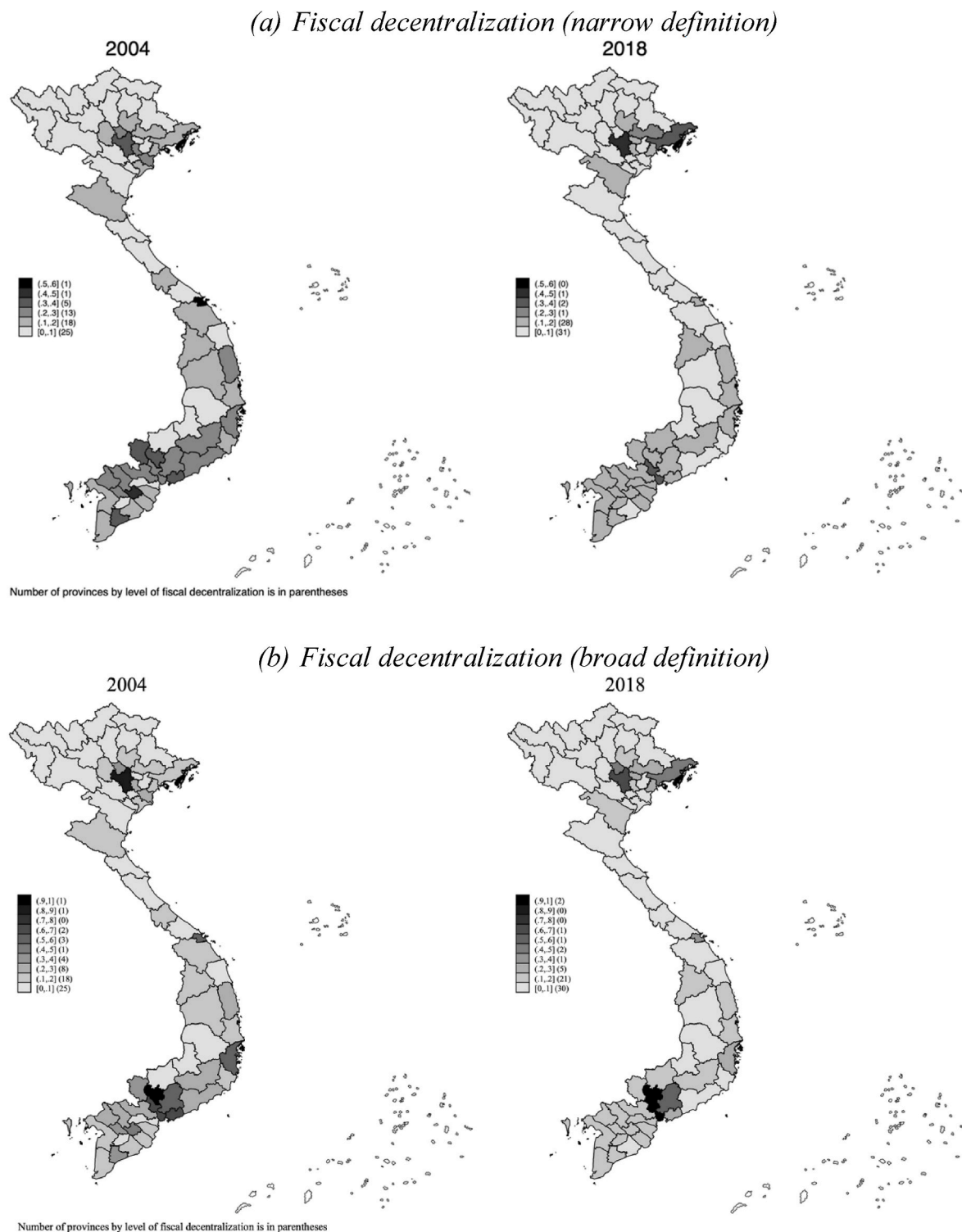


Fig. 1. Provincial fiscal decentralization level in 2004–2018

Notes: This figure shows cross-province differences in fiscal decentralization in Vietnam for the years (a) 2004 and (b) 2018. Darker areas indicate higher levels of fiscal decentralization, reflecting the proportion of revenue that provincial governments can retain relative to their total assigned spending. Fiscal decentralization, by the broad definition, is classified within a wider range of 0–1 compared to 0–0.6 under the narrow definition. The maps highlight changes in the patterns of fiscal decentralization between 2004 and 2018, with increased decentralization in the Red River Delta contrasted by decreases in other regions.

both local and household characteristics. Since fiscal decentralization is measured at the provincial level, it is likely to correlate with unobserved local-level characteristics. In Equation (6), the unobserved factors μ_{jt} can be separated into temporal and provincial components. In this specification, we control for provincial fixed effects to eliminate time-invariant

unobserved factors at the provincial level. However, the regression model may still produce bias due to endogeneity if fiscal decentralization correlates with time-variant factors.

We use an identification strategy by leveraging an exogenous source of variation in fiscal decentralization, given regional heterogeneity and

Table 1
Summary statistics.

	N	Mean	SD	Min	Max
Panel A. Household-level variables					
Electricity spending in last 12 months (thousand USD)	72251	0.087	0.120	0	15.043
Access to national power grid	73816	0.966	0.182	0	1
Energy poverty: Ten-percent rate (TPR) index	73816	0.239	0.426	0	1
Energy poverty: Half-median (M2) index	73816	0.201	0.401	0	1
Energy poverty: Low-income high-cost (LIHC) index	73816	0.050	0.218	0	1
Energy poverty: Limited modern appliances (MOT) index	73816	0.284	0.451	0	1
Energy poverty: Multidimensional energy poverty (MEPI) index	73816	0.169	0.226	0	1
Age of household head	73816	50.104	13.942	11	105
Gender of household head (1 = Male, 0 = Female)	73816	0.753	0.431	0	1
High school graduation of household head [1 = yes, 0 = No]	73816	0.334	0.472	0	1
Number of household members	73816	4.092	1.650	1	20
Minority or not [1 = yes, 0 = No]	73816	0.166	0.372	0	1
Urban area or not [1 = yes, 0 = No]	73816	0.278	0.448	0	1
Living size (square meter)	73743	73.701	47.388	4	2010
Rental or not [1 = yes, 0 = No]	73743	0.016	0.127	0	1
Number of electrical machines	73816	5.984	4.087	0	84
Panel B. Province-level variables					
Fiscal decentralization – narrow definition (FD1) [0–1]	504	0.128	0.091	0.001	0.613
Fiscal decentralization – broad definition (FD2) [0–1]	504	0.196	0.224	0.001	0.987
GDP growth [0–1]	504	0.195	0.151	−0.228	1.776
Industrial output (thousand USD per capita)	504	0.656	1.328	0.032	15.493
Population density (million people per square km)	504	0.221	0.213	0.034	1.127
Inflation rate	504	108.5	6.185	99.53	135.45
Public investment (thousand USD per capita)	504	0.211	0.166	0.020	1.18
Private investment (thousand USD per capita)	504	0.327	0.259	0.018	1.705
Foreign direct investment (FDI) (thousand USD per capita)	493	0.118	0.254	0.000	2.584

Notes: This table reports the summary characteristics of the data used in this study. We collect household-level data from the Vietnam Household Living Standards Survey (VHLSS) for the years 2004–2018. Provincial-level data are obtained from the annual statistical yearbooks published by the General Statistics Office (GSO).

the spillover effects of economic activities across neighboring provinces that influence both economic activities and local revenue. We calculate a jack-knifed average of fiscal decentralization in neighboring provinces

and use it as an instrument for the fiscal decentralization of a province in a given year. Due to the spillover effects of economic activities, the areas surrounding leading economic regions often experience greater economic activity. For example, industrial zones are being established in the neighboring areas of special and centrally controlled municipalities like Ho Chi Minh City and Hanoi. Also, the central government typically allocates revenue and plans economic development strategies by region in Vietnam. Close geographical proximity promotes the cross-border flow of labor, investment, infrastructure, and technology, thereby generating spatial dependencies in provincial government revenue systems. Moreover, economic activities and local revenues can associate across neighboring provinces through peer pressure and imitation effects.⁴ For each province i in a given year t , IV_{it} is derived from the leave-one-out average of fiscal decentralization of neighboring provinces within the same region, excluding the observation for province i itself. The instrument is calculated as:

$$IV_{it} = \frac{1}{N_r - 1} \sum_{k \neq j} FD_{kt} \quad (7)$$

where N_r is the number of provinces in region r and k denotes the neighboring provinces of province j . A key assumption for the use of IV_{it} is that it causes no direct impact on energy poverty outcomes of a province other than through its impact on the level of fiscal decentralization in province j . Using regional variations in fiscal decentralization by both narrow and broad definitions, we expect this instrument not to affect energy poverty in a province directly, as it does not impact public investment in that province, e.g., investment in infrastructure related to manufacturing and living standards that influence energy demand and energy poverty at the provincial level. Besides, to address endogeneity concerns partially induced by unobserved factors, the premise for the exogeneity of this instrument is that many unobserved macroeconomic factors that correlate with energy poverty, public revenue, and expenditure growth are province-specific. Therefore, while the structural economic transformation of neighboring provinces significantly predicts the fiscal decentralization of a province, this relationship remains independent of unobserved province-specific characteristics.

Building on the 2FE and IV strategies, we address concerns about endogeneity related to potential unobserved confounders for fiscal decentralization at the provincial level. These approaches generate consistent estimations of the parameters even in the presence of province-specific effects that might bias the estimate of δ . Next, endogeneity issues from reverse causality may challenge the inference. The energy supply capabilities of a province might explain energy demand and economic activities, which in turn affect public revenue. In an energy-poor province, the government may also allocate more resources to infrastructure in order to increase economic efficiency. To account for the dynamic and persistent relationship between energy poverty and fiscal decentralization, we employ a dynamic panel data model with a lagged dependent variable included on the right-hand side of the model. The dynamic model is specified as follows:

$$\begin{aligned} \text{Energy poverty}_{jt} = & \text{Energy poverty}_{jt-1}\omega + \text{Fiscal decentralization}_{jt}\delta \\ & + P_{jt}\gamma + \lambda_j + \kappa_t + \mu_{jt} \end{aligned} \quad (8)$$

We employ the two-step system GMM with lagged levels and differences in explanatory variables as instruments to mitigate the endogeneity from simultaneity (Blundell and Bond, 1998). In the first step, the province-specific fixed effects λ_j are removed through differencing:

⁴ The logic is further supported by recent studies on fiscal decentralization (e.g., the survey of Martinez-Vazquez et al. (2017)).

$$\Delta \text{Energy poverty}_{jt} = \Delta \text{Energy poverty}_{jt-1} \omega + \Delta \text{Fiscal decentralization}_{jt} \delta + \Delta P_{jt} \gamma + \Delta \mu_{jt} \quad (9)$$

where Δ denotes the first differences. Lagged values of the explanatory variables are used as instruments for their first differences in the two-step system GMM estimator, resolving the challenge of finding valid external instruments. In the second stage, residuals from the first stage are used to construct an optimal weighting matrix. This matrix adjusts for heteroscedasticity and inconsistencies in the error structure to make the estimators robust. Hence, the two-step system GMM provides efficient and unbiased parameter estimates, particularly with dynamic panel data where traditional methods may fail to control for endogeneity or autocorrelation (Blundell and Bond, 2023).

5. Findings

Table 2 presents the OLS estimates of the effect of narrow fiscal decentralization (FD1) on different measures of energy poverty at the household level. Across the five energy poverty indicators, the estimated coefficients of FD1 are negative in the range -0.020 to -0.142 . More specifically, for TPR, M2, and MOT, the coefficients are statistically significant at the 1 % level, with slopes of between -0.087 and -0.142 , indicating a relatively strong association between higher fiscal decentralization and lower energy poverty. On average, a one standard deviation increase in fiscal decentralization is associated with a 0.7 to 1.3 percentage-point decrease in the probability of household energy poverty in one of the dimensions of income, expenditure, or energy quality. For all dimensions of energy poverty, the coefficient for MEPI is smaller at -0.033 but remains statistically significant at the 5 % level. When fiscal decentralization increases by one standard deviation, the

likelihood of households experiencing energy poverty reduces by an average of 0.3 percentage points. The estimated coefficient of the impact of FD1 on LIHC is modest and not precisely estimated, with the slope not statistically significant at conventional levels.

In term of household characteristic, age is positively associated with measures of energy poverty, while male household heads and those living in a larger household tend to have lower energy poverty levels. Minorities appear to face higher energy poverty across indicators, suggesting possible socio-economic disparities in access to energy services. Estimated coefficients for rental status, household living size, and the number of electrical appliances are as expected and confirm the role of household economic conditions in determining energy poverty outcomes. All specifications include province and year-fixed effects to control for unobserved time-invariant provincial characteristics and macro-level effects.

Compared to the narrow definition in Table 2, the broad definition of fiscal decentralization (FD2) in Table 3 generally exhibits smaller coefficient magnitudes on household-level energy poverty measures. Under the broad definition, the estimated effects on TPR, M2, and MOT remain negative and statistically significant. The coefficient in the LIHC model remains negative yet statistically insignificant, as shown in Table 2. However, the slope in the MEPI model is no longer statistically significant, as shown in Table 3. The coefficient of FD2 is statistically significant at the 1 % level for three measures (TPR, M2, and MOT), with coefficients ranging from approximately -0.047 to -0.064 . These results suggest that broad fiscal decentralization still shows an overall negative association with household energy poverty, although the estimated slope is somewhat smaller compared to the more focused measure of fiscal decentralization by the narrow definition. To some extent, these differences imply that narrower measures of fiscal decentralization may capture more direct aspects of decentralization of fiscal authority that

Table 2

The impact of fiscal decentralization (narrow definition) on energy poverty: Household level estimates.

	(1)	(2)	(3)	(4)	(5)
	TPR	M2	LIHC	MOT	MEPI
FD1	-0.142^{***} (0.029)	-0.087^{***} (0.025)	-0.020 (0.015)	-0.128^{***} (0.027)	-0.033^{**} (0.013)
Household characteristics					
Age	0.003^{***} (0.000)	0.001^{***} (0.000)	0.001^{***} (0.000)	0.002^{***} (0.000)	0.002^{***} (0.000)
Gender	-0.055^{***} (0.004)	-0.047^{***} (0.003)	-0.030^{***} (0.002)	-0.037^{***} (0.003)	-0.038^{***} (0.002)
High school graduation	0.006^* (0.003)	0.007^{***} (0.003)	0.005^{***} (0.002)	0.024^{***} (0.003)	0.009^{***} (0.002)
Number of household members	-0.056^{***} (0.001)	-0.031^{***} (0.001)	-0.014^{***} (0.001)	-0.010^{***} (0.001)	-0.024^{***} (0.000)
Minority	0.040^{***} (0.005)	0.240^{***} (0.005)	0.055^{***} (0.003)	0.063^{***} (0.005)	0.094^{***} (0.003)
Urban area	-0.002 (0.004)	-0.064^{***} (0.003)	-0.014^{***} (0.002)	-0.020^{***} (0.004)	-0.024^{***} (0.002)
Living size	-0.001^{***} (0.000)	-0.000^{***} (0.000)	-0.000^{***} (0.000)	-0.000^{***} (0.000)	-0.000^{***} (0.000)
Rental	-0.065^{***} (0.012)	-0.013 (0.010)	-0.039^{***} (0.006)	0.009 (0.011)	-0.028^{***} (0.006)
Number of electrical machines	-0.021^{***} (0.000)	-0.029^{***} (0.000)	-0.009^{***} (0.000)	-0.057^{***} (0.000)	-0.023^{***} (0.000)
Constant	0.221^{***} (0.082)	0.538^{***} (0.071)	0.063 (0.043)	0.488^{***} (0.077)	0.282^{***} (0.038)
Observations	72403	72403	72403	72403	72403
Province FE	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes
Provincial controls	Yes	Yes	Yes	Yes	Yes

Notes: This table presents OLS estimates of the effect of fiscal decentralization, using the narrow definition, on energy poverty at the household level. Dependent variables are different measures of energy poverty, including the ten-percent rate (TPR), half-median (M2), low-income high-cost (LIHC), limited modern appliances (MOT), and multidimensional energy poverty (MEPI) index. Provincial characteristic control variables include GDP growth, industrial output, population density, inflation, public investment, private investment, and FDI inflow. Besides household-level control variables from the VHLSS, the model includes region dummies (the North Central Coast region is the base group) and year dummies (VHLSS, 2004 is the base group). Standard errors are clustered at the provincial level and are reported in parentheses ***p < 0.01, **p < 0.05, *p < 0.10.

Table 3

The impact of fiscal decentralization (broad definition) on energy poverty: Household level estimates.

	(1)	(2)	(3)	(4)	(5)
	TPR	M2	LIHC	MOT	MEPI
FD2	−0.058*** (0.021)	−0.047*** (0.018)	−0.005 (0.011)	−0.064*** (0.020)	−0.011 (0.010)
Household characteristics					
Age	0.003*** (0.000)	0.001*** (0.000)	0.001*** (0.000)	0.002*** (0.000)	0.002*** (0.000)
Gender	−0.055*** (0.004)	−0.047*** (0.003)	−0.030*** (0.002)	−0.037*** (0.003)	−0.038*** (0.002)
High school graduation	0.006* (0.003)	0.007*** (0.003)	0.005*** (0.002)	0.024*** (0.003)	0.009*** (0.002)
Number of household members	−0.056*** (0.001)	−0.031*** (0.001)	−0.014*** (0.001)	−0.011*** (0.001)	−0.024*** (0.000)
Minority	−0.040*** (0.005)	−0.240*** (0.005)	0.055*** (0.003)	0.063*** (0.005)	0.094*** (0.003)
Urban area	−0.002 (0.004)	−0.064*** (0.003)	−0.014*** (0.002)	−0.020*** (0.004)	−0.024*** (0.002)
Living size	−0.001*** (0.000)	−0.000*** (0.000)	−0.000*** (0.000)	−0.000*** (0.000)	−0.000*** (0.000)
Rental	−0.065*** (0.012)	−0.014 (0.010)	−0.039*** (0.006)	0.009 (0.011)	−0.028*** (0.006)
Number of electrical machines	−0.021*** (0.000)	−0.029*** (0.000)	−0.009*** (0.000)	−0.057*** (0.000)	−0.023*** (0.000)
Constant	0.221*** (0.082)	0.538*** (0.071)	0.063 (0.043)	0.488*** (0.077)	0.282*** (0.038)
Observations	72403	72403	72403	72403	72403
Province FE	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes
Provincial controls	Yes	Yes	Yes	Yes	Yes

Notes: This table presents OLS estimates of the effect of fiscal decentralization, using the broad definition, on energy poverty at the household level. Dependent variables are different measures of energy poverty, including the ten-percent rate (TPR), half-median (M2), low-income high-cost (LIHC), limited modern appliances (MOT), and multidimensional energy poverty (MEPI) index. Provincial characteristic control variables include GDP growth, industrial output, population density, inflation, public investment, private investment, and FDI inflow. Besides household-level control variables from the VHLSS, the model includes region dummies (the North Central Coast region is the base group) and year dummies (VHLSS, 2004 is the base group). Standard errors are clustered at the provincial level and are reported in parentheses ***p < 0.01, **p < 0.05, *p < 0.10.

affect household energy circumstances, e.g., local governments are likely to use fully retained revenue for local infrastructure that helps to reduce energy poverty.⁵

Table 4 reports 2FE estimates of the effect of fiscal decentralization on energy poverty at the provincial level, using both a narrow definition in columns 1–5 and a broad definition in columns 6–10. Compared to the household-level estimates using cross-sectional estimates, this specification takes advantage of longitudinal variation across provinces, allowing for control of time-invariant unobserved provincial heterogeneity. By including province and year-fixed effects and clustering standard errors at the provincial level, the analysis isolates the impact of fiscal decentralization on provincial averages of energy poverty measures. Using the narrow definition with FD1, the estimated coefficients on TPR, M2, LIHC, and MEPI are all negative and statistically significant at the 1 % level, with magnitudes ranging from around −0.032 to −0.171. The results are consistent with those in Table 2, suggesting that a one standard deviation increase in the narrow fiscal decentralization index corresponds to a 0.3–1.6 reduction in these measures of energy poverty. Meanwhile, the impact on MOT is negative yet smaller in magnitude and is not statistically different from zero. Under the broad definition FD2, in Columns 6–10, the effects on TPR, M2, LIHC, and MEPI also remain negative and statistically significant. However, the

coefficient estimates are generally smaller in magnitude than those in Columns 1–5. For instance, the estimated slopes on TPR of −0.043 and on M2 of −0.113 are somewhat attenuated relative to the narrow definition. As in the narrow definition models, MOT again shows a statistically insignificant coefficient. These provincial-level panel findings confirm the impact of fiscal decentralization, whether narrowly or broadly defined, in lowering energy poverty.

For other control variables, GDP growth has a negative impact on energy poverty across most indicators, indicating the importance of economic growth in reducing energy poverty. Industrial output has a positive impact on energy poverty, particularly in models that use a broad definition. Population density yields mixed results with significant reductions in TPR, and inflation generally worsens energy poverty. The impacts of public and private investment are consistently significant, with private investment showing the largest positive impact. FDI has marginally positive coefficients but is not statistically significant.

Table 5 presents IV estimates of the impact of fiscal decentralization on energy poverty at the provincial level. This specification addresses potential endogeneity by instrumenting each province's decentralization level with the jack-knifed average of neighboring provinces' decentralization. Panel B shows strong first-stage results in that the instrument has a positive and statistically significant effect on fiscal decentralization. The first-stage F-statistic proposed by [Olea and Pflueger \(2013\)](#) tests weak instrument validity, even when data exhibit non-homoscedasticity, serial correlation, or clustering problems ([Andrews et al., 2019](#)). In this analysis, first-stage F-statistics above 80 and a significant Wald test based on [Kleibergen and Paap \(2006\)](#) indicate that the instrument is robust and relevant. Turning to the second-stage estimates in Panel A, the coefficients of fiscal decentralization remain negative for most energy poverty indicators, with statistical significance at the 5 % or 1 % level. Again, the effects are

⁵ To check robustness, whether different energy poverty indicators are good proxies and the impact of fiscal decentralization on the energy aspect is consistent, we estimate its effects on electricity spending and access in Appendix Table A1. The results show that fiscal decentralization has no impact on electricity spending, so price or income effects do not bias the findings. However, the impact of fiscal decentralization on energy consumption is likely driven by infrastructure investment, thereby improving household electricity access.

Table 4

The impact of fiscal decentralization on energy poverty: Provincial-level estimates using fixed effects.

	Fiscal decentralization 1 (narrow definition)					Fiscal decentralization 2 (broad definition)				
	<u>TPR</u>	<u>M2</u>	<u>LIHC</u>	<u>MOT</u>	<u>MEPI</u>	<u>TPR</u>	<u>M2</u>	<u>LIHC</u>	<u>MOT</u>	<u>MEPI</u>
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
FD	−0.074*** (0.019)	−0.171*** (0.025)	−0.032*** (0.009)	−0.022 (0.102)	−0.055*** (0.019)	−0.043*** (0.011)	−0.113*** (0.021)	−0.022*** (0.008)	−0.014 (0.082)	−0.037** (0.015)
GDP growth	−0.051*** (0.011)	−0.012 (0.011)	−0.022*** (0.004)	−0.302*** (0.043)	−0.068*** (0.008)	−0.048*** (0.011)	−0.014 (0.011)	−0.022*** (0.004)	−0.302*** (0.043)	−0.069*** (0.008)
Industrial output	0.002 (0.001)	0.005 (0.003)	0.003** (0.001)	0.005 (0.112)	0.004 (0.002)	0.004** (0.001)	0.009*** (0.003)	0.003*** (0.001)	0.005 (0.012)	0.005** (0.002)
Pop density	−0.050*** (0.011)	0.102 (0.082)	0.007 (0.031)	0.097 (0.331)	0.031 (0.062)	−0.035*** (0.013)	0.116 (0.084)	0.009 (0.031)	0.098 (0.332)	0.036 (0.063)
Inflation	−0.002*** (0.000)	−0.002*** (0.000)	−0.001*** (0.000)	0.002 (0.001)	−0.001*** (0.000)	−0.002*** (0.000)	−0.002*** (0.000)	−0.001*** (0.000)	0.002 (0.001)	−0.001*** (0.000)
Public investment	−0.011 (0.011)	0.113*** (0.013)	0.026*** (0.005)	−0.003 (0.054)	0.037*** (0.010)	−0.012 (0.011)	0.112*** (0.014)	0.025*** (0.005)	−0.003 (0.054)	0.036*** (0.010)
Private investment	0.131*** (0.008)	0.024** (0.010)	0.062*** (0.004)	0.021 (0.039)	0.064*** (0.007)	0.130*** (0.008)	0.023** (0.010)	0.062*** (0.004)	0.021 (0.039)	0.063*** (0.007)
FDI	0.004 (0.007)	0.011 (0.009)	0.005 (0.003)	0.005 (0.035)	0.006 (0.007)	0.005 (0.007)	0.007 (0.009)	0.004 (0.003)	0.005 (0.035)	0.005 (0.007)
Constant	0.434*** (0.031)	0.347*** (0.032)	0.105*** (0.012)	0.132 (0.128)	0.226*** (0.024)	0.431*** (0.031)	0.343*** (0.032)	0.104*** (0.012)	0.131 (0.128)	0.225*** (0.024)
Observations	493	493	493	493	493	493	493	493	493	493

Notes: This table presents panel estimates of the effect of fiscal decentralization, using both the narrow (columns 1–5) and broad definitions (columns 6–10), on energy poverty at the province level. Dependent variables are provincial averages of different measures of energy poverty, including the ten-percent rate (TPR), half-median (M2), low-income high-cost (LIHC), limited modern appliances (MOT), and multidimensional energy poverty (MEPI) index. All estimates include province- and year-fixed effects. Standard errors are clustered at the province level and are reported in parentheses ***p < 0.01, **p < 0.05, *p < 0.10.

Table 5

The impact of fiscal decentralization on energy poverty: Provincial-level estimates using IV.

	Fiscal decentralization 1 (narrow definition)					Fiscal decentralization 2 (broad definition)				
	<u>TPR</u>	<u>M2</u>	<u>LIHC</u>	<u>MOT</u>	<u>MEPI</u>	<u>TPR</u>	<u>M2</u>	<u>LIHC</u>	<u>MOT</u>	<u>MEPI</u>
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Panel A. Second-stage estimates. Dependent variables are energy poverty indicators										
FD	−0.103** (0.044)	−0.355*** (0.048)	−0.110*** (0.018)	−0.070 (0.151)	−0.159*** (0.032)	−0.085** (0.037)	−0.133*** (0.036)	−0.054*** (0.014)	−0.044 (0.124)	−0.061** (0.025)
GDP growth	−0.050*** (0.011)	−0.013 (0.012)	−0.025*** (0.004)	−0.278*** (0.038)	−0.068*** (0.008)	−0.043*** (0.012)	−0.006 (0.012)	−0.021*** (0.005)	−0.284*** (0.041)	−0.064*** (0.008)
Industrial output	0.002* (0.001)	0.002 (0.002)	0.001** (0.001)	0.002 (0.005)	0.002* (0.001)	0.006** (0.002)	0.006** (0.002)	0.003*** (0.001)	−0.001 (0.008)	0.003** (0.002)
Pop density	−0.044*** (0.014)	0.045*** (0.015)	−0.005 (0.005)	0.000 (0.047)	0.001 (0.010)	−0.006 (0.028)	0.068** (0.027)	0.011 (0.011)	−0.044 (0.093)	0.013 (0.019)
Inflation	−0.002*** (0.000)	−0.001*** (0.000)	−0.001*** (0.000)	0.002 (0.001)	−0.001*** (0.000)	−0.002*** (0.000)	−0.002*** (0.000)	−0.001*** (0.000)	0.001 (0.001)	−0.001*** (0.000)
Public investment	−0.012 (0.011)	0.056*** (0.012)	0.009** (0.004)	0.005 (0.037)	0.016** (0.008)	−0.013 (0.011)	0.058*** (0.011)	0.009** (0.004)	0.007 (0.037)	0.017** (0.008)
Private investment	0.132*** (0.008)	0.044*** (0.009)	0.053*** (0.003)	0.027 (0.028)	0.059*** (0.006)	0.131*** (0.008)	0.036*** (0.008)	0.051*** (0.003)	0.024 (0.028)	0.055*** (0.006)
FDI	0.004 (0.007)	0.010 (0.008)	0.004 (0.003)	0.017 (0.025)	0.007 (0.005)	0.006 (0.008)	0.013* (0.007)	0.005* (0.003)	0.015 (0.026)	0.009* (0.005)
Constant	0.430*** (0.031)	0.370*** (0.033)	0.136*** (0.012)	0.161 (0.105)	0.261*** (0.022)	0.419*** (0.032)	0.378*** (0.032)	0.135*** (0.013)	0.181* (0.109)	0.264*** (0.022)
Panel B. First-stage estimates. The dependent variable is FD										
IV	0.725*** (0.071)	0.725*** (0.071)	0.725*** (0.071)	0.725*** (0.071)	0.725*** (0.071)	0.410*** (0.035)	0.410*** (0.035)	0.410*** (0.035)	0.410*** (0.035)	0.410*** (0.035)
Observations	493	493	493	493	493	493	493	493	493	493
First-stage F-stat	83.74	83.74	83.74	83.74	83.74	81.60	81.60	81.60	81.60	81.60
Cragg-Donald weak identification test	101.62	101.62	101.62	101.62	101.62	46.28	46.28	46.28	46.28	46.28
Kleibergen-Paap Wald Stat.	34.31	34.31	34.31	34.31	34.31	18.83	18.83	18.83	18.83	18.83

Notes: This table presents IV estimates of the effect of fiscal decentralization, using both the narrow (columns 1–5) and broad definitions (columns 6–10), on energy poverty at the province level. Dependent variables are provincial averages of different measures of energy poverty, including the ten-percent rate (TPR), half-median (M2), low-income high-cost (LIHC), limited modern appliances (MOT), and multidimensional energy poverty (MEPI) index. *Olea and Pflueger's (2013)* effective first-stage F-statistic provides evidence that the instruments used are strong. In contrast, the Kleibergen-Paap Wald test tests the null hypothesis that the instrument is weakly correlated with the endogenous regressor. To reject this null hypothesis, Stock and Yogo's critical values are applied, which are: 16.38 for a 10 % maximal IV size, 8.96 for 15 %, 6.66 for 20 %, and 5.53 for 25 % maximal IV size. Standard errors are clustered at the province level and are reported in parentheses ***p < 0.01, **p < 0.05, *p < 0.10.

precisely estimated at conventional significance levels, except for Columns 4 and 9 for the MOT index. For the narrow definition, the slopes range from around -0.103 to -0.355 . The broad definition in Columns 6–10 results in similarly negative coefficients of -0.061 to -0.133 .

Compared with the 2FE results, the magnitudes of the fiscal decentralization coefficients in the IV models are typically larger. The FE estimates may be biased toward zero if there is a positive correlation between unobserved determinants of energy poverty and fiscal decentralization. That is, in the 2FE models, any simultaneity bias where provinces that tend to have higher levels of energy poverty also make different fiscal choices could weaken the estimated coefficients. The estimated coefficients, while inflated, increase toward the true causal magnitude. One limitation of IV estimates is that using regional variation in fiscal decentralization as the source of exogenous variation may reduce overall variation in fiscal decentralization across observations. Using regional average fiscal decentralization levels smooths out many individual provincial-level variations. In other words, the IV approach estimates the impact of fiscal decentralization as a local average treatment effect (LATE) where decentralization levels are influenced by the regional variation leveraged by the instrument. Thus, the estimated coefficients may not fully generalize to all provinces where the variation in fiscal decentralization is less dependent on regional factors.

Table 6 provides the results of the two-step system of GMM estimates. This approach extends beyond standard 2FE estimates by estimating the dynamic relationship between energy poverty and fiscal decentralization. Panel A reports the short-run coefficients, showing that fiscal decentralization has a consistently negative impact on all

provincial energy poverty indicators with statistical significance at the 1 % level. Panel B presents the long-run coefficients, where the long-run impact of fiscal decentralization on energy poverty is calculated by dividing the estimated short-run coefficients by $(1 - \omega)$, where ω represents the coefficient of the lagged energy poverty. The long-run effects in Panel B show that the magnitude of the impact of decentralization increases substantially when the cumulative influence is taken into account. For instance, while the short-run effect under the narrow definition may be around -0.027 to -0.272 , the long-run coefficient is more than -0.234 .

Fig. 2 provides the estimated effects of decentralization on energy poverty measures across five major regions of Vietnam: the Red River Delta, Northeast-Northwest, Central, Southeast, and Mekong River Delta, using the narrow definition. Several patterns emerge. First, while most regions experience negative coefficients, indicating that greater decentralization helps reduce energy poverty, the magnitude and significance levels vary depending on both the region and the specific energy poverty measure. Unlike other regions, the Southeast often shows positive coefficients. In contrast, the Northeast-Northwest region appears to benefit the most from fiscal decentralization, with significant reductions in energy poverty. The Central, Red River Delta, and Mekong River Delta regions are less responsive to decentralization, with smaller or statistically insignificant coefficients.

It is common practice in Vietnam to categorize household income into five quantiles when assessing poverty by different income groups. Besides financial strain, lower-income households often have limited access to modern energy services because economic poverty is closely

Table 6

The impact of fiscal decentralization on energy poverty: Provincial-level estimates using two-step system GMM.

	Fiscal decentralization 1 (narrow definition)					Fiscal decentralization 2 (broad definition)				
	TPR	M2	LIHC	MOT	MEPI	TPR	M2	LIHC	MOT	MEPI
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Panel A. Short-run coefficients										
Lag.	0.922*** (0.003)	0.678*** (0.005)	0.953*** (0.004)	0.566*** (0.004)	0.633*** (0.003)	0.923*** (0.002)	0.739*** (0.004)	0.949*** (0.004)	0.558*** (0.004)	0.607*** (0.002)
FD	-0.132*** (0.004)	-0.091*** (0.002)	-0.027*** (0.001)	-0.272*** (0.016)	-0.086*** (0.003)	-0.058*** (0.002)	-0.035*** (0.000)	-0.008*** (0.000)	-0.196*** (0.008)	-0.056*** (0.002)
GDP growth	-0.029*** (0.001)	0.035*** (0.001)	-0.018*** (0.000)	-0.313*** (0.005)	-0.100*** (0.001)	-0.025*** (0.001)	0.040*** (0.001)	-0.017*** (0.000)	-0.333*** (0.008)	-0.025*** (0.001)
Industrial output	0.001*** (0.000)	-0.001*** (0.000)	0.000*** (0.000)	-0.004*** (0.000)	0.001*** (0.000)	0.003*** (0.000)	0.000 (0.000)	0.000*** (0.000)	-0.013*** (0.001)	0.003*** (0.000)
Pop density	-0.018*** (0.004)	0.033*** (0.005)	0.003*** (0.001)	0.120*** (0.022)	-0.005** (0.002)	0.006 (0.005)	0.047*** (0.005)	0.006*** (0.001)	0.003 (0.018)	0.006 (0.005)
Inflation	0.003*** (0.000)	-0.001*** (0.000)	0.000*** (0.000)	-0.018*** (0.000)	-0.001*** (0.000)	0.003*** (0.000)	-0.001*** (0.000)	0.000*** (0.000)	-0.017*** (0.000)	0.003*** (0.000)
Public investment	0.040*** (0.003)	0.057*** (0.002)	0.011*** (0.001)	-0.051*** (0.012)	-0.003 (0.002)	0.036*** (0.002)	0.052*** (0.002)	0.011*** (0.001)	-0.024 (0.015)	0.036*** (0.002)
Private investment	0.069*** (0.001)	-0.040*** (0.001)	0.002*** (0.000)	-0.338*** (0.010)	0.022*** (0.001)	0.065*** (0.001)	-0.046*** (0.001)	0.000 (0.000)	-0.327*** (0.010)	0.065*** (0.001)
FDI	0.003*** (0.001)	0.003*** (0.001)	0.001*** (0.000)	0.030*** (0.005)	0.005*** (0.001)	0.004*** (0.001)	0.003*** (0.001)	0.002*** (0.000)	0.029*** (0.005)	0.004*** (0.001)
Constant	0.401*** (0.008)	0.130*** (0.002)	-0.019*** (0.001)	2.691*** (0.048)	0.261*** (0.002)	0.395*** (0.005)	0.115*** (0.002)	-0.020*** (0.001)	2.637*** (0.059)	0.395*** (0.005)
Panel B. Long-run coefficients										
FD	-1.320*** (0.017)	-0.282*** (0.007)	-0.585*** (0.056)	-0.626*** (0.122)	-0.234*** (0.005)	-0.753*** (0.019)	-0.123*** (0.005)	-0.313*** (0.040)	-0.443*** (0.012)	-0.142*** (0.003)
Observations	431	431	431	431	431	431	431	431	431	431
No. groups	63	63	63	63	63	63	63	63	63	63
No. IVs	72	72	72	72	72	72	72	72	72	72
AR[1]	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
AR[2]	0.372	0.555	0.456	0.481	0.137	0.737	0.695	0.514	0.342	0.122
Hansen p-value	0.588	0.587	0.588	0.195	0.640	0.586	0.586	0.587	0.205	0.633

Notes: The two-step system GMM estimator, using lagged (1–2) instruments for the endogenous variables, is employed to derive the estimates. To assess the validity of the system GMM results, several diagnostic tests are conducted. The Hansen test checks the null hypothesis that the instruments are uncorrelated with the error term. Moreover, the AR[1] and AR[2] tests are used to examine first- and second-order autocorrelation in the first-differenced residuals. Long-run coefficients are calculated by dividing the estimated short-run coefficients of the explanatory variables β by $(1 - \omega)$, where γ represents the coefficient of the lagged dependent variable. Standard errors are reported in parentheses. * $p < 0.10$; ** $p < 0.05$; *** $p < 0.01$.

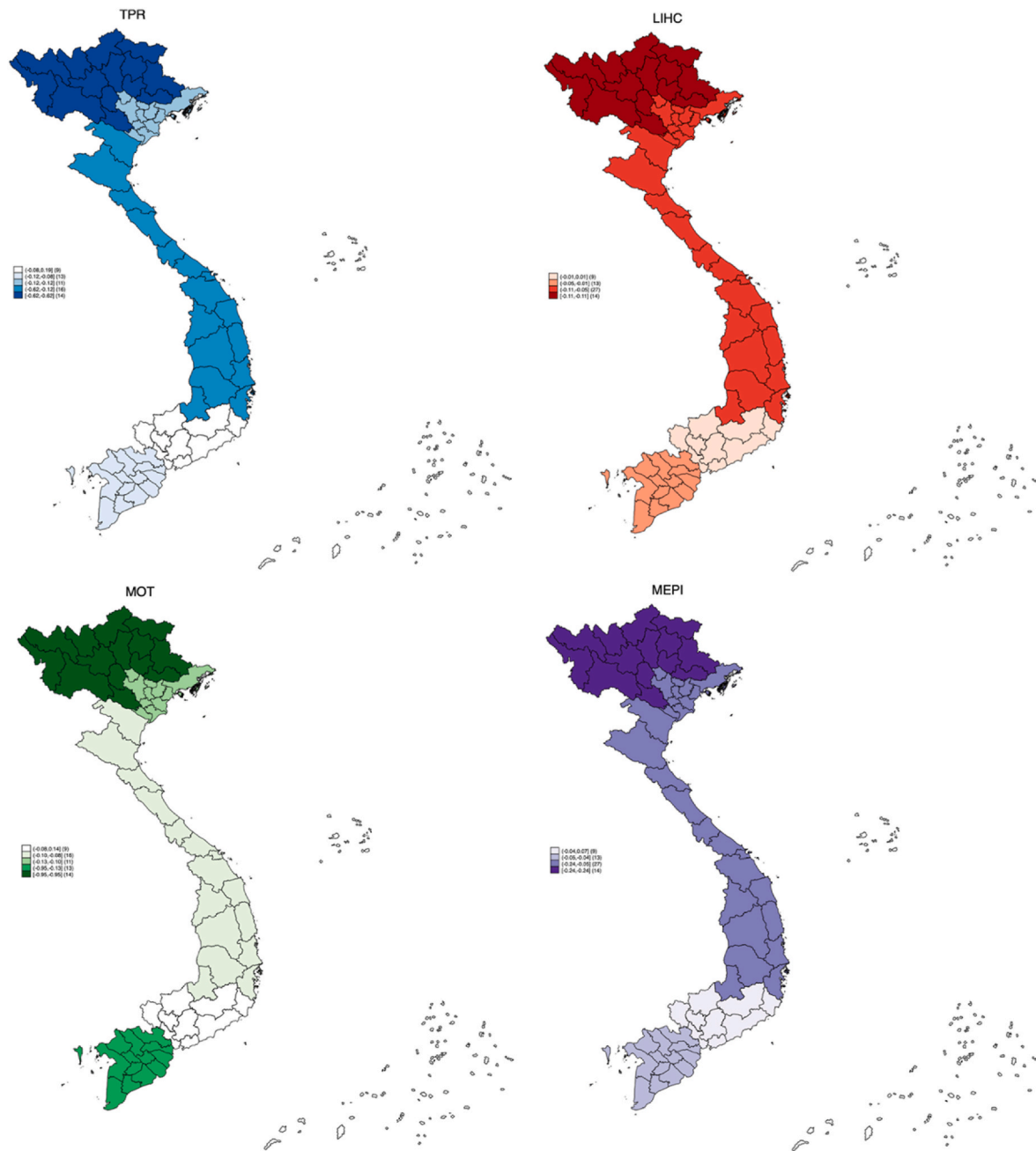


Fig. 2. Estimated impact of fiscal decentralization (narrow definition) on energy poverty by region

Notes: This figure presents the estimated impact of fiscal decentralization (narrow definition) on different indicators of energy poverty by subsamples classified by five regions in Vietnam: Red River Delta, Northeast and Northwest, Central, Southeast, and Mekong River Delta regions. Since the estimated coefficients of fiscal decentralization across different regions are driven by the definition of energy poverty, we do not report the estimated coefficients for the M2 index by region. The point estimates and 95 % confidence intervals are presented in Appendix Table A5. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

related to energy poverty. Fig. 3 plots the estimated effects of narrow fiscal decentralization on four household energy poverty indicators (TPR, M2, MOT, and MEPI) across these five income quantiles. The negative estimated slopes in the lower-income quantiles (1–3) tend to be larger in magnitude, whilst the estimates for higher-income quantiles (4–5) are close to zero. This result implies that fiscal decentralization has a particularly significant effect on reducing energy poverty among poor households.

6. Discussion

Our study shows that fiscal decentralization can significantly reduce energy poverty in a developing country. This finding is consistent with recent studies that reveal the important role of fiscal policy in tackling energy poverty. However, most of the previous studies focus on government spending (e.g., [Nguyen and Su \(2022\)](#)). This study goes further by showing that fiscal decentralization is an important policy tool for this purpose. The findings are consistent with recent studies, such as [Xia et al. \(2022\)](#) and [Wang et al. \(2023\)](#) in the case of China. Our findings

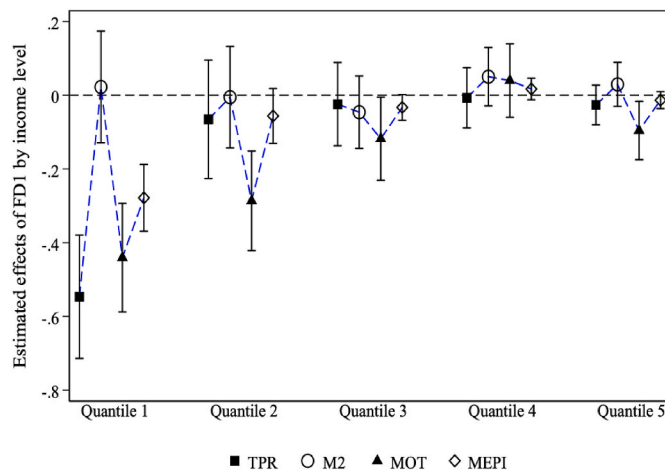


Fig. 3. Estimated impact of fiscal decentralization (narrow definition) on energy poverty by income level

Notes: This figure illustrates the point estimates with 95 % confidence intervals (CIs) of the estimated impact of fiscal decentralization (narrow definition) on different indicators of energy poverty across five quantiles of income. Given that the computation of the LIHC index is mostly driven by income quantiles, we do not report the estimated coefficients for this index by income quantile, as the estimated coefficients for quantiles 4 and 5 are missing.

add fresh evidence for a small developing country, Vietnam. Going further, the findings can be seen as strong evidence to support the first-generation theory of decentralization (Dinh et al., 2023), where fiscal decentralization can enhance local government competition through the mechanism of “voting with their feet” (Tiebout, 1956), improving public goods delivery (Oates, 1985; Tiebout, 1956).

Interestingly, the finding can be considered a novel extension linking the literature on fiscal decentralization to energy poverty. As discussed, studies of fiscal decentralization in the case of Vietnam show that fiscal decentralization can promote better local governance (e.g., Dinh et al. (2023)), while it is a positive stimulator for both social and economic factors (e.g., Nguyen (2017); Do and Park (2022); Diem and Hart (2022)). As such, fiscal decentralization can make a positive contribution to inclusive development (Bui et al., 2023). Meanwhile, previous studies show that fiscal decentralization improves health outcomes (Dao and Hong, 2023) and reduces inequality (Diem and Hart, 2022). Our findings indicate that it can also play a role in sustainable development by reducing energy poverty. These results are significant in that they inform government policies related to sustainable development and highlight the relevance of designing fiscal reforms with institutional support and equitable resource distribution mechanisms to achieve a broader impact.

Additionally, the findings suggest that this effect is more pronounced among lower-income households and in regions such as the Northeast-Northwest. It should be noted that fiscal decentralization in Vietnam does not appear to be uniformly distributed across all regions (Thanh and Canh, 2019). For example, areas with higher levels of economic development, such as the Southeast—which includes major cities like Ho Chi Minh City and highly industrialized provinces like Binh Duong and Dong Nai—or the capital city of Hanoi in the Red River Delta, tend to have greater fiscal revenue capacity. Provinces in the Northeast-Northwest, Central, and Central Highland regions often rely on fiscal transfers from the central government to support their expenditure. As a result, these provinces are less reliant on collecting tax revenues and may focus on improving local governance and public service provision. In comparison, provinces with greater fiscal autonomy—such as Ho Chi Minh City, which does not receive central fiscal transfers and must meet tax collection requirements—might follow the norm of the “race to the bottom” in fiscal decentralization theory. This pattern has been observed in Vietnam (Thanh and Canh, 2019). Such

dynamics may contribute to a more pronounced effect of fiscal decentralization on reducing energy poverty in poorer regions and among lower-income households.

7. Concluding remarks

This paper studies how fiscal decentralization affects household energy poverty using different cross-sectional and panel data models. The results show that greater fiscal decentralization is consistently associated with lower energy poverty across energy poverty indicators. In both household-level and province-level models, we find that an increase in fiscal decentralization reduces the probability of energy poverty, with effects that are more significant and larger in magnitude under the narrow definition. For the robustness check, the IV estimates model confirms the main findings from the OLS estimates. Furthermore, dynamic panel estimations using two-step system GMM suggest that fiscal decentralization has both short-term and long-term effects in reducing energy poverty, with the long-run coefficients being larger and accumulating over time. The results also show regional and income heterogeneity. In the regional analysis, regions like the Northeast-Northwest benefit more from decentralization, while others, such as the Southeast, show a less consistent impact. In the income analysis, the effects of decentralization are larger for low-income households, suggesting that fiscal decentralization might have a redistributive role when combined with pro-poor spending policies and local service provision for basic needs, such as energy access.

These findings have policy implications. *First*, we recommend that fiscal decentralization is a good candidate for governments in developing countries to use in achieving sustainable development, as our findings show that fiscal decentralization can reduce energy poverty. Meanwhile, there is evidence that fiscal decentralization can improve governance and promote socioeconomic development. That is, fiscal decentralization can help governments achieve dual benefits. In fact, fiscal decentralization has been used in policy reform in developing countries such as China and India (Li and Xu, 2023; Mishra et al., 2023), but this policy should be further studied and adopted in other developing countries. *Second*, our findings suggest that the implementation of fiscal decentralization in the context of sustainable development must incorporate the perspective of energy poverty and other social welfare. Specifically, energy policies, including green energy transition, energy accessibility, and energy affordability, must be among the principal responsibilities of local governments during fiscal decentralization. Their inclusion would help limit “race to the bottom” behavior among local governments, where they focus on tax collection rather than improving public service delivery and taking care of social welfare.

Our findings show that fiscal policy reforms can reduce energy poverty and promote equitable energy access, but some limitations remain. The benchmark estimates do not include all time-variant factors that may influence how decentralization affects energy poverty, such as changes in the quality of local governance or household consumption preferences. To deal with the challenge in specification, the IV strategy uses regional average decentralization as an instrument, and that may reduce variation across provinces. Future studies might consider identifying a quasi-experiment in fiscal policy as a premise for developing a proper identification strategy.

CRedit authorship contribution statement

Nguyen Doan: Writing – review & editing, Writing – original draft, Validation, Methodology, Formal analysis, Data curation, Conceptualization. **Canh Phuc Nguyen:** Writing – review & editing, Writing – original draft, Validation, Methodology, Conceptualization. **Thanh Dinh Su:** Writing – review & editing, Writing – original draft, Validation, Conceptualization.

Declaration of generative AI in scientific writing

We do not use generative AI in scientific writing upon submission of the paper.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.enpol.2025.114845>.

Data availability

Data will be made available on request.

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